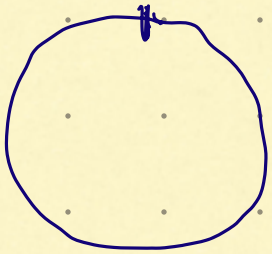
"Adiabatic" computation



$$y = f(x)$$

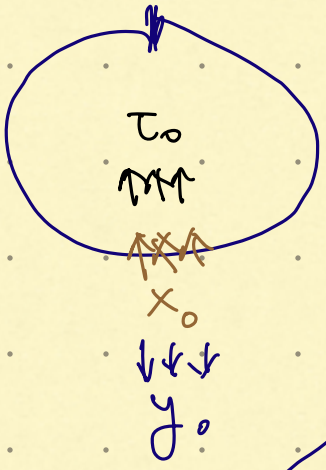
$$y_0 = f(x_0)$$



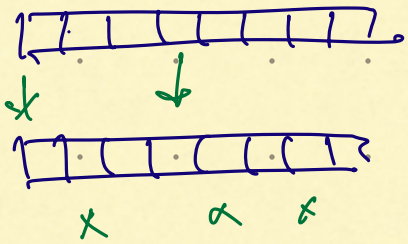
NAND



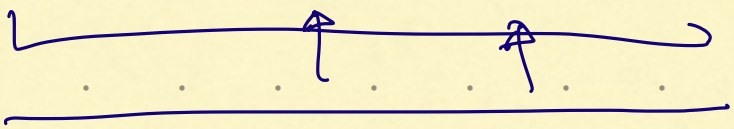
$f \leftarrow$ NAND, NAND, NAND | AND, OR, XOR



$f(x')$



$$x' = \delta(x' | x_0) \oplus x_0$$



$\delta = 01001010$

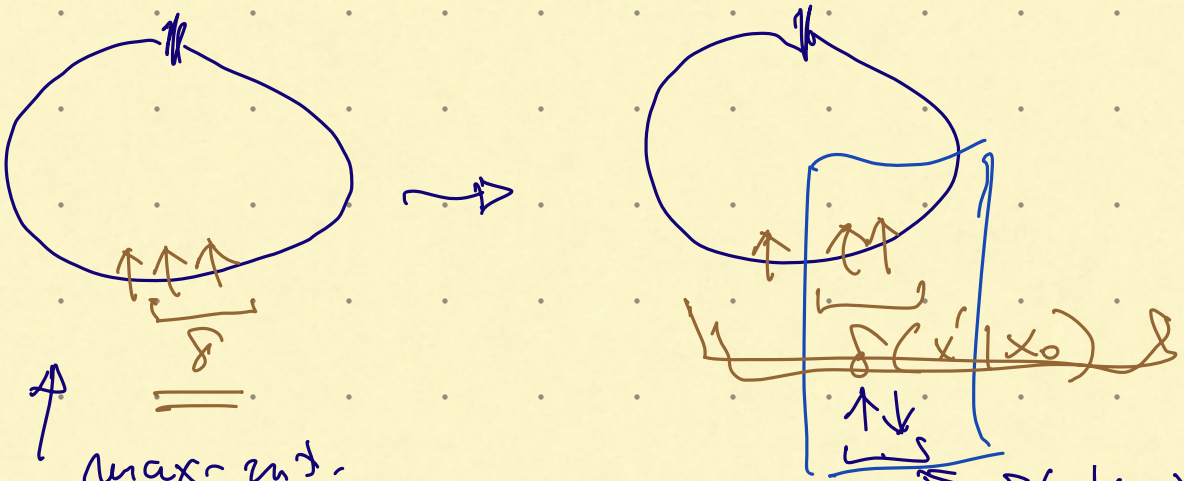
7 6 5 4 3 2 1 0

$\oplus =$

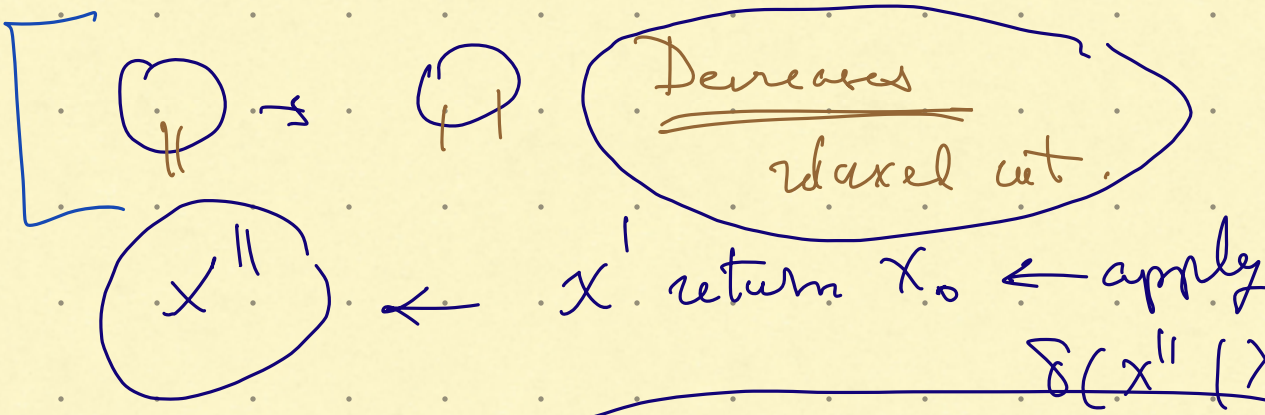
x_1	x_2	y
1	1	0
1	0	1
0	1	1
0	0	0

δ

$$f(x') = f(\delta(x'|x_0) \oplus x_0)$$



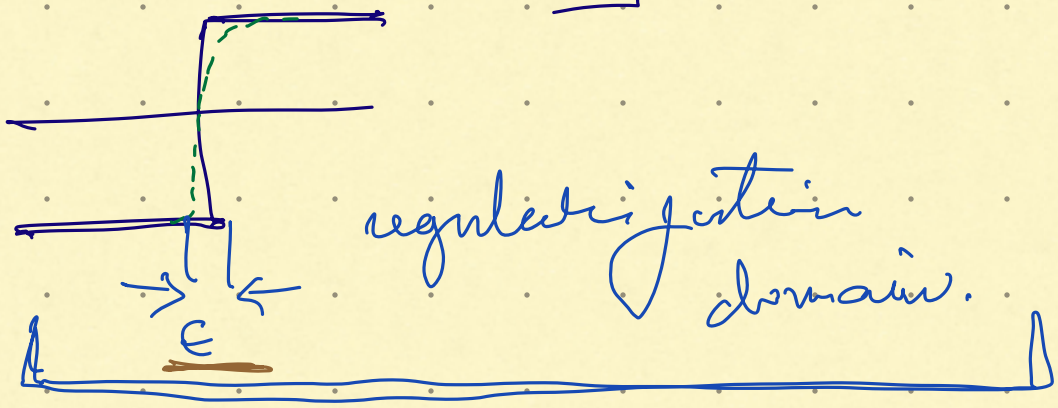
$$y' = \delta(y'|y_0) \oplus y_0$$



$$C(\psi) = C(\sigma) + \frac{1}{\gamma} \sum A_{mn} \sigma_m \sigma_n |x_m - x_n|$$

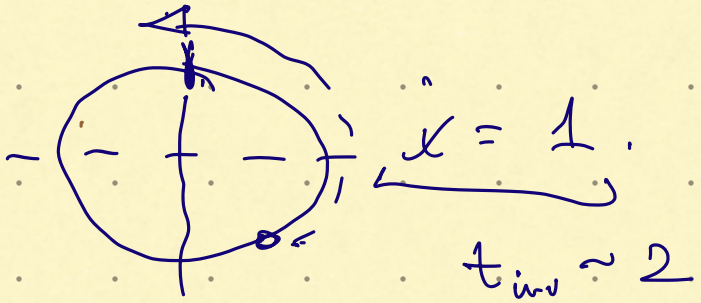
$$\dot{x}_m = \sigma_m \sum_n A_{mn} \sigma_n \operatorname{sign}(x_m - x_n)$$

[Filippov framework
Clarke sub-differentials
chambers...]



Input displacement rate: v

$x_m(t) = x_m(0) + vt$, $m \in$ { displaced input spins }.



$v \sim \frac{\epsilon}{2}$ slowly for reg. $U \sim 2$.

$v >> \frac{\epsilon}{2}$ → problems w/ convergence

$v \sim \frac{\epsilon}{2}$ Maybe not

Noise may help.

Energy budget of computation

Landauer limit.

$$\Delta E \geq k_B T \ln 2$$

Reversible computing

10^9 bits to comp.

$\Delta E \sim 10^9 \cdot k_B T \ln 2$.

V_2 model.

$$\Delta E = N_{Bound} k_B T \ln 2.$$

N_{Bound} ← input bits.

FPS. Counter Strike. Call of Duty.

2 hours. ΔE < 1 Watt per hour.

Highest settings.

10 GB of assets.

10^4 by pixels. + 10 GB of what?

$20 \cdot 10^9 \cdot 10 \cdot 10^{-23} \cdot 300 \cdot 2$

$2 \cdot 10^{13} \cdot 10^{-23} \cdot 2 \sim 4 \cdot 10^{-10}$ Joules.

1 KWh = 3600000 ~ $3.6 \cdot 10^6$ Joules.

$$\Delta E = \Delta E^{(0)} + \mathcal{O}\left(\frac{1}{T}\right).$$